Tailoring GitHub Copilot

From building blocks to team-wide distribution

GitHub Copilot
Dev Days
Napoli

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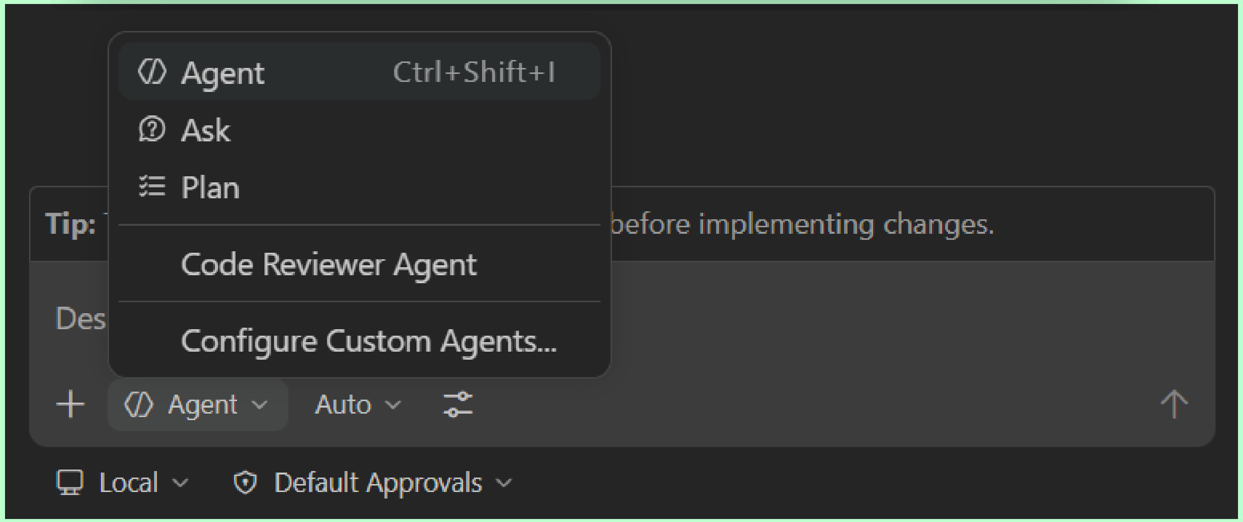
Agents

.github/agents/reviewer-agent.json

```
---
name: Code Review Agent
tools: [execute/runTests, read, agent, search, todo]
model: GPT-5.4 (copilot)
handoffs:
  - label: Start fixes
    agent: csharp-developer
    prompt: Implement the fixes
    model: GPT-5.4 (copilot)
---
Review the changes in this PR from different angles.
Perform these reviews in parallel:

- Run the security-reviewer agent to check for vulnerabilities
- Run the performance-reviewer agent to identify bottlenecks
- Run the accessibility-reviewer agent to verify a11y compliance

Consolidate findings into a single review summary.
```



Subagents are specialized AI assistants that handle side tasks in their own context, returning only a summary, keeping your main conversation clean.

Each runs with its own context window, system prompt, tools, and permissions.

Standard

Skills

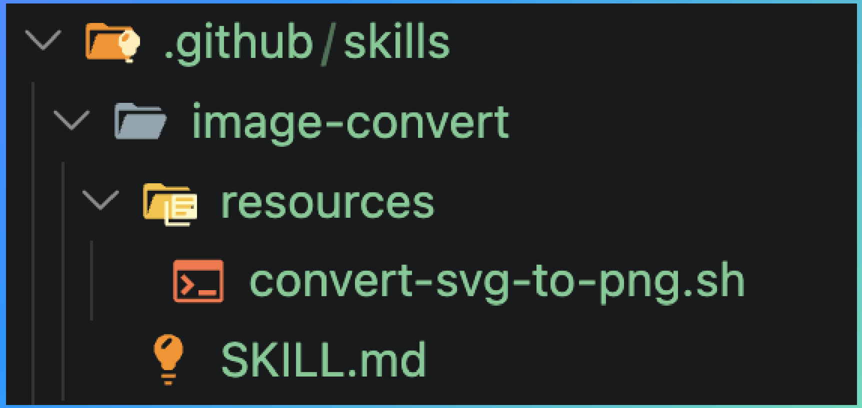
A Skill (Agent Skill) is a self-contained folder plus optional scripts and resources that gives Copilot a specialized, portable capability.

It loads automatically when the prompt matches its description.

.github/skills/image-convert/SKILL.MD

```
---
name: image-convert
description: Converts SVG images to PNG format.
Use when asked to convert SVG files.
allowed-tools: shell
---

When asked to convert an SVG to PNG, run the
`convert-svg-to-png.sh` script from this skill's
base directory, passing the input SVG file path as the
first argument.
```



Standard

MCP Server

An MCP server (Model Context Protocol) is a process that exposes tools, resources, and prompts to AI agents over a standardized protocol. It's how Copilot reaches external systems like databases, APIs, file systems, third-party services, without bespoke integrations.

VS Code acts as the MCP client: the server runs locally or remotely and is configured per-workspace or globally. Once connected, its tools become available to agents alongside built-in ones.

.copilot/mcp-config.json

```json
{
  "mcpServers": {
    "playwright": {
      "type": "local",
      "command": "npx",
      "args": ["@playwright/mcp@latest"],
      "env": {},
      "tools": ["*"]
    },
    "context7": {
      "type": "http",
      "url": "https://mcp.context7.com/mcp",
      "headers": {
        "CONTEXT7_API_KEY": "YOUR-API-KEY"
      },
      "tools": ["*"]
    }
  }
}
```

Hooks

Hooks are user-defined shell commands, HTTP endpoints, or LLM prompts that execute automatically at specific points in GitHub Copilot's lifecycle.

.github/hooks/hooks.json

```
"userPromptSubmitted": [
  {
    "type": "command",
    "bash": "./scripts/log-prompt.sh",
    "powershell": "./scripts/log-prompt.ps1",
    "cwd": "scripts",
    "env": {
      "LOG_LEVEL": "INFO"
    }
  }
]
```

Preview

Availability hooks for Copilot

- sessionStart
- userPromptSubmit
- preToolUse
- postToolUse
- preCompact
- subagentStart
- subagentStop
- stop

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Availability hooks for Copilot CLI

- sessionStart
- sessionEnd
- userPromptSubmitted
- preToolUse
- postToolUse
- errorOccurred

The distribution problem

Teams build extensions like agents, skills, MCP servers, hooks.

But sharing them across the organisation is hard: capabilities get rebuilt locally, best practices stay isolated, and value doesn’t scale.

plugin.json

```json
{
  "name": "dev-tools",
  "description": "Blazor development utilities",
  "version": "1.0.0",
  "author": {
    "name": "Gerardo Greco",
    "email": "gerardo.greco@blexin.com"
  },
  "license": "MIT",
  "keywords": ["blazor", "frontend"],
  "agents": "agents/",
  "skills": ["skills/", "extra-skills/"],
  "hooks": "hooks.json",
  "mcpServers": ".mcp.json"
}
```

Plugins

Plugins let you extend GitHub Copilot with extensions that can be shared across projects and teams.

Create a (private) marketplace

A marketplace is a catalog of plugins that someone else has created and shared.

.github/plugin/marketplace.json

```json
{
  "name": "blexin-marketplace",
  "owner": {
    "name": "Blexin srl"
  },
  "metadata": {
    "description": "Curated plugins for our team",
    "version": "1.0.0"
  },
  "plugins": [
    {
      "name": "dev-tools",
      "source": "./plugins/dev-tools"
    }
  ]
}
```

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Demo

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References

https://docs.github.com/en/copilot/reference/hooks-configuration

https://code.visualstudio.com/docs/copilot/customization/hooks

https://code.visualstudio.com/docs/copilot/customization/custom-agents

https://docs.github.com/en/copilot/how-tos/copilot-cli/customize-copilot/add-skills

https://docs.github.com/en/copilot/concepts/context/mcp

https://docs.github.com/en/copilot/how-tos/copilot-cli/customize-copilot/plugins-creating

https://docs.github.com/en/copilot/how-tos/copilot-cli/customize-copilot/plugins-marketplace

https://github.com/render93/gh-copilot-dev-days-2026

Thank you

gh.io/copilot